

Sajeevan Veeriah

Robotics, Mechatronics, AI/ML & End-To-End Automation Engineer

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PROFILE

I engineer complete systems across the physical, control, digital and operational boundary, then verify them as one working system. My practice spans robotics, industrial automation, embedded systems, industrial IT/OT, AI/ML and engineering software from requirements through integration, commissioning and handover.

ENGINEERING CAPABILITY

- **Industrial IT, OT and automation:** PLC and HMI/SCADA integration, MES and production-data interfaces, field devices, drives, industrial networking, FAT, SAT, commissioning and traceable handover.
- **Robotics and mechatronics:** ROS 2 Humble, Nav2, MoveIt 2, Gazebo Fortress, SLAM, EKF sensor fusion, localisation, planning, motion control and physical prototyping.
- **Embedded and connectivity:** ESP32, STM32, FreeRTOS, C/C++, PCB bring-up, CAN, UART, I2C, SPI, BLE, LoRaWAN, MQTT and Linux services.
- **AI/ML, data and software:** Python, scikit-learn, OpenCV, YOLO, time-series analysis, MATLAB, Simulink, TypeScript, React, Rust, SQL, CI/CD and local-first software.
- **Verification and delivery:** requirements, instrumentation, calibrated testing, regression, fault isolation, data integrity, traceable QA, release boundaries and technical handover.

SELECTED ENGINEERING EVIDENCE

Autonomous Navigation Rover on ROS 2

2024-2025 | Hardware build with simulation-validated autonomy

Built the differential-drive platform and integrated LiDAR, IMU, SLAM, EKF state estimation, Nav2 planning, control and recovery across independently testable ROS 2 nodes. Gazebo Fortress regression runs and RViz inspection checked maps, transforms, pose and planned paths.

Boundary: No fleet-deployment or certified functional-safety claim.

ESP32 Clinical Ataxia Assessment Device

2025 | Honours capstone and assessed embedded prototype

Designed the device, custom PCB and enclosure, implemented deterministic 100 Hz acquisition across four Hall-effect channels, Bluetooth live display and CSV/PDF reporting, then checked accuracy, direction, reversal, drift and temperature behaviour against reference instruments in MATLAB.

Boundary: Not a certified medical device and not a claim of clinical efficacy.

SWL Pricing and Inventory Control

2026 | Client-commissioned system, release 1.2.0

Sole engineer across requirements, deterministic pricing and matching rules, React/TypeScript interfaces, Tauri/Rust desktop packaging, SQLite, file contracts, CI and release evidence. More than 500 automated checks cover unit, property, browser, accessibility, desktop and Rust behaviour.

Boundary: Production installation, signing, automatic updates and rollout remain outside the release boundary.

Open Industrial Automation

2026 | Public Phase 1 reference platform

Built a deterministic industrial reference process with an operator HMI, Engineering Studio, typed project model, traceability and verification gates. The platform makes the control, information, cybersecurity and evidence boundaries explicit.

Boundary: A reference and simulation platform, not certified safety control or site-commissioned plant automation.

PROFESSIONAL EXPERIENCE

Automation and Controls Engineer | Process automation systems integrator

Jan 2026 - Jun 2026 | Regulated pharmaceutical, biotechnology and food production

- Delivered PLC logic, HMI/SCADA, field-device, drive, MES and production-data integration under GMP controls.
- Migrated validated application content from iFIX to PVI+, completed functional checks, and supported FAT, SAT, commissioning, handover and mobile-robot fault tracing.

PROFESSIONAL EXPERIENCE CONTINUED

Product Development Test Engineer (Contract) | Global automotive OEM via engineering consultancy

Oct 2025 - Jan 2026 | Vehicle software integration and ADAS product development

- Validated vehicle-software integration and ADAS behaviour through feature-vehicle, breadboard and regression testing.
- Instrumented test vehicles and used Vector CANoe and CANalyzer to capture CAN and CAN FD evidence for fault isolation and software-readiness verification.

Technical Officer & Quality Assurance | ABMARC

Jul 2024 - Aug 2025 | Automotive testing, energy, emissions and compliance engineering

- Delivered ADR and EURO emissions work, EV/PHEV range testing and vehicle-systems evaluation using calibrated instrumentation, data acquisition and CAN tools.
- Maintained traceable QA, safety, regulatory and test records for certification review and audit.

Consultant Engineer, IoT and Projects Administrator | DuxTel

Feb 2024 - Aug 2024 | Networking, low-power telemetry and remote asset visibility

- Built field-to-dashboard solutions with ESP32, sensors, LoRaWAN AU915, MQTT, ChirpStack, InfluxDB and Grafana.
- Supported CAN and GPS capture, custom PCB work, MikroTik connectivity and Linux services from field trial through handover.

Production Line Team Lead & Cellar Hand | IDL Australia

Aug 2022 - Feb 2024 | Beverage manufacturing, packaging and traceable production

- Led day-to-day output, changeovers and shift handovers while maintaining batch quality, traceability and safe operation.
- Performed first-response fault recovery and supported installation and commissioning checks during canning-line upgrades.

Production Line Operator and Commissioning Support | Carbon Revolution

Sep 2021 - Dec 2021 | Carbon-fibre wheel production and Industry 4.0 quality control

- Operated production equipment, recorded in-process quality and traceability evidence, and supported equipment trials, setup and first-level recovery.

Undergraduate Quality Engineer | Thornton Engineering Australia

Mar 2018 - Mar 2020 | Structural steel and heavy fabrication

- Supported drawing review, inspection and test plans, manufacturing data records, material traceability and AS/NZS compliance across design, workshop and inspection boundaries.

EDUCATION AND PROFESSIONAL STANDING

Bachelor of Mechatronics Engineering (Honours), Distinction | Deakin University | 2025

Higher National Diploma in Mechatronics, Robotics and Automation Engineering, Distinction | Cardiff Metropolitan University | 2016

Member, Engineers Australia | Current Victorian driver licence

TRAINING AND LANGUAGES

Training: Lean Six Sigma Foundation, JIRA and Agile, KAIZEN, Industrial Automation and IIoT, AI/ML and CAD.

Languages: English, Tamil and Sinhala.

WORKING STYLE

Understand the whole system, isolate faults and leave clear evidence for verification, commissioning and handover. Beyond project delivery, I play and help run my cricket club, mentor fellow students, and maintain personal robotics and hardware builds.